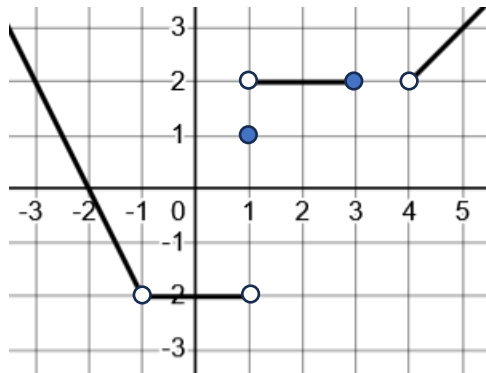


Teaching Set 1

0). This problem is about making sure you understand how to **read** a piecewise function and graph.

a). Big idea: When (in what kind of situation) would you need to use a piecewise defined function? Why?

Consider this piece-wise defined function.



$$f(x) = \begin{cases} -2x - 4 & \text{for } x < -1 \\ -2 & \text{for } -1 < x < 1 \\ 1 & \text{for } x = 1 \\ 2 & \text{for } 1 < x \leq 3 \\ (x - 4) + 2 & \text{for } x > 4 \end{cases}$$


 y = x, shifted right 4 and up 2

b). Why does the dashed interval $-1 < x < 1$ use $<$ instead of $\leq$ signs? How do you know when to use $<$ and when $\leq$?

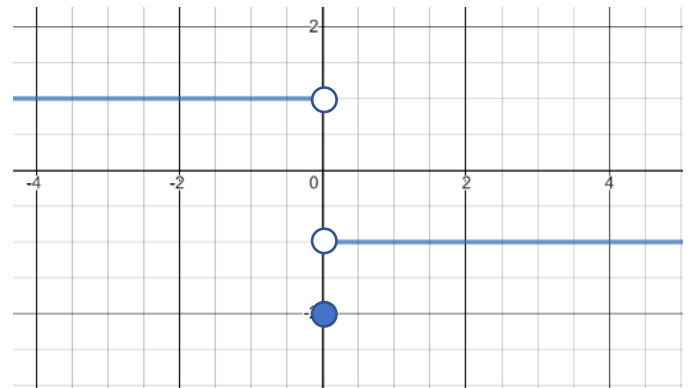
c). What part of the graph does the 3rd piece: “1 for $x = 1$ ” correspond to?

d). Find $f(4.5)$, $f(1)$ and $f(-1.5)$ using the graph. How could you find them using the equation only?

1). This is about **writing a piecewise equation** for a very simple piecewise function.

Write a piecewise defined function corresponding to the following graph.

Note that your function should have 3 pieces and its domain should be all real numbers.

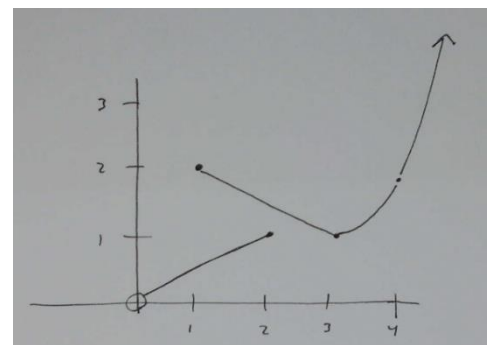


2). **Clear up common confusions.** There is one major thing wrong with how each of these piecewise functions is written. For each, write 1 sentence of English saying what the problem is.

a). What’s wrong with this piece-wise graph?

b). There’s nothing *mathematically* wrong with the equation for $f(x)$, but this person didn’t write their idea down in the right way. What’s wrong with it? What *should* it look like?

c). What’s mathematically wrong with the equation $g(x)$?



$$f(x) = \begin{cases} y = -x & \text{if } x < 0 \\ y = x & \text{if } x \geq 0 \end{cases}$$

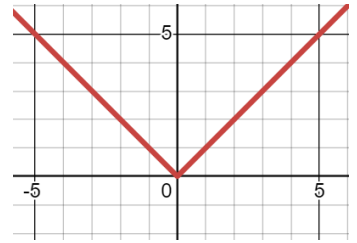
$$g(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

3). a). Here's a graph of $y = |x|$.

Write a piece-wise defined function for it **without using** $|x|$. In other words, imagine it as two separate lines.

b). Create a piecewise defined function for $f(x) = |x + 3|$ that uses 2 linear equations. (sketch the graph if that helps)

c). Double check that your equation works: What *should* $f(-3)$ be? Does your function give the right output? If not, re-think it!



4). Let $f(x) = \begin{cases} x^2 + 1, & x \leq 0 \\ \frac{1}{x}, & x > 0 \end{cases}$

a. What's $f(-2)$?

b. What's $f(2)$?

c. What is $f(f(-3))$?

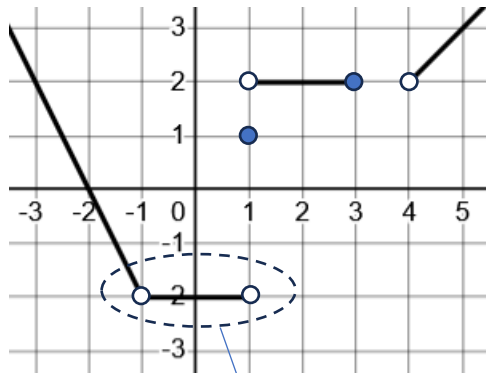
d. Sketch a graph of $y = f(x)$

Teaching Set 1 - Solutions.

0). a). Big idea: When (in what kind of situation) would you need to use a piecewise defined function? Why?

If you have a situation where different equations apply for different input ranges. This happens a lot for human rules, like determining prices based on how much you're buying, but it also happens a lot in science. [*expand on this...*]

Consider this piece-wise defined function.



$$f(x) = \begin{cases} -2x - 4 & \text{for } x < -1 \\ -2 & \text{for } -1 < x < 1 \\ 1 & \text{for } x = 1 \\ 2 & \text{for } 1 < x \leq 3 \\ (x - 4) + 2 & \text{for } x > 4 \end{cases}$$

$y = x$, shifted right 4 and up 2

a). Why does the dashed interval $-1 < x < 1$ use $<$ instead of $\leq$ signs? How do you know when to use $<$ and when $\leq$?

You can see **open circles** at $x = -1$ and $x = 1$. That means there is no value for the function at that point. If we'd said $-1 \leq x \leq 1$ we would be saying that we could use that piece of the equation if $x = -1$ or $x = 1$, in which case the output would be 2. Just think whether you want that piece of your equation to apply to the x -values at the end-point(s). If you do, use a closed circle in the graph and a $<$ or $>$ sign. If you don't, use an open circle in the graph, and a $\leq$ or $\geq$ sign.

b). What part of the graph does the 3rd piece: "**1 for $x = 1$** " correspond to?

Just the single dot at $(1, 1)$. We're saying the output is 1 when $x = 1$, which is just a single point.

c). Find $f(4.5)$, $f(1)$ and $f(-1.5)$ using the graph. How could you find them using the equation only?

Using the graph, find $x = 4.5$ and look at the graph to find the y -value at that x -value.

In this case $f(4.5) = 2.5$, $f(1) = 1$, and $f(-1.5) = -1$

To use the equation, normally you'd plug in the x -value to find the y -value. But which equation? You need to find which of the intervals on the right contains 4.5 or 1 or -1.5 and then you can use *that* part of the equation to plug in.

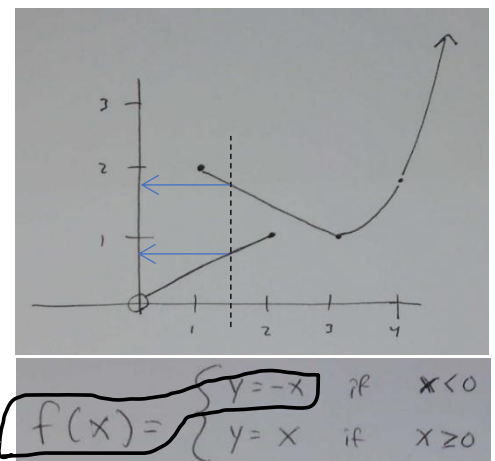
$$1. f(x) = \begin{cases} 1 & , x < 0 \\ -2 & , x = 0 \\ -1 & , x > 0 \end{cases}$$

2). Clear up common confusions.

a). What's wrong with this piece-wise graph? For many inputs, the graph is showing **two outputs** (for example at $x = 1.5$ shown).

b). There's nothing *mathematically* wrong with the equation for $f(x)$, but this person didn't write their idea down in the right way. What's wrong with it? What *should* it look like? They shouldn't have put the $y =$ inside the brackets.

It's redundant, since we already have the $f(x) =$ on the outside. You wouldn't write $f(x) = y = -x$. You'd just write $f(x) = -x$.



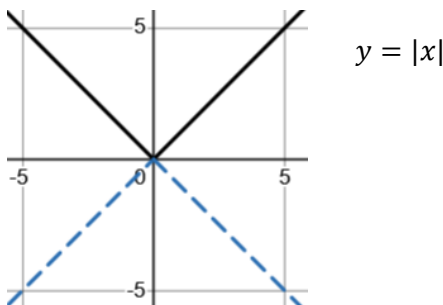
c). What's mathematically wrong with the equation $g(x)$? **Think about what this says about $g(0)$. What is the y-value when $x = 0$? If you look at the intervals, the first interval says that if $x \leq 0$ the output is 1. So seems like $g(0) = 1$. But the second interval says that if $x \geq 0$ the output is -1. So that part claims $g(0) = -1$.**

$$g(x) = \begin{cases} 1 & \text{if } x \leq 0 \\ -1 & \text{if } x \geq 0 \end{cases}$$

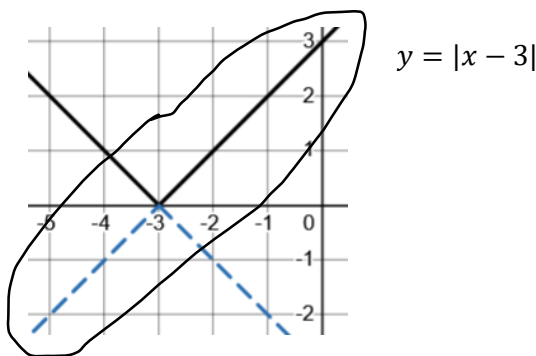
You can't have $x = 0$ included in both intervals! That causes the function to have two different outputs at $x = 0$, which isn't what we want for functions. (We want just one output for each input).

3. It helps to sketch the graph. If you're going to imagine the shape as made up of lines, you could sketch in the rest of the line to help you think.

a). $f(x) = \begin{cases} -x & x \leq 0 \\ x & x > 0 \end{cases}$



b). $f(x) = \begin{cases} -x - 3 & x \leq -3 \\ x + 3 & x > -3 \end{cases}$



To get this right, imagine one line at a time.

For example, the one that's circled. It's $y = x + 3$

But we only want it to apply when $x > -3$

4). Let $f(x) = \begin{cases} x^2 + 1, & x \leq 0 \\ \frac{1}{x}, & x > 0 \end{cases}$

a. What's $f(-2)$?

b. What's $f(2)$?

c. What is $f(f(-3))$?

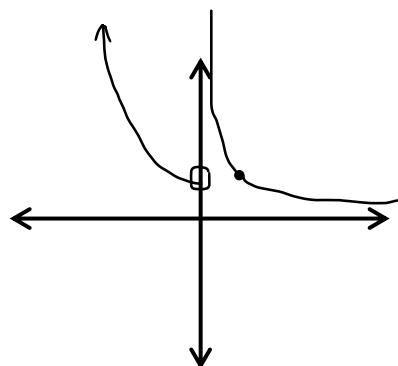
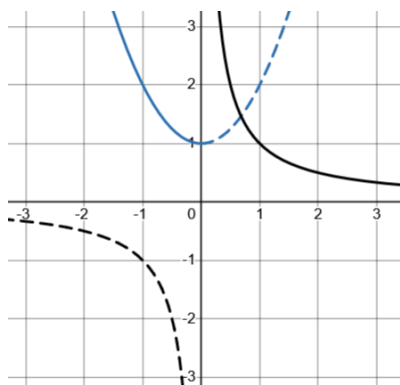
The whole trick here is to choose the right equation to plug into.

For $f(-2)$ we want to find y when $x = -2$. That fits the first interval: $x \leq 0$. So plug into $x^2 + 1$ to get $f(-2) = 5$.

For $f(2)$ we want to find y when $x = 2$. That fits the second interval: $x > 0$. So plug into $\frac{1}{x}$ to get $f(2) = \frac{1}{2}$.

d. Sketch a graph of $y = f(x)$. If you want to, imagine each equation separately. Erase the parts you don't want.

This is NOT the answer.
But it can help you
think about the parts.



Piecewise Defined Functions – Teaching Set 2

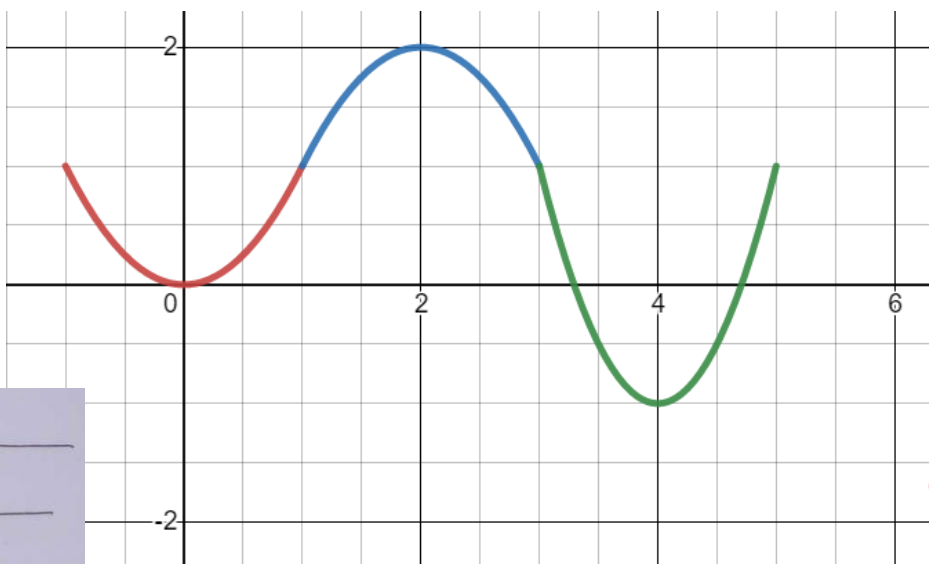
1). Let's make a pattern out of transformed copies of...

$g(x) = x^2$. Here it is to the right!

Complete this **piece-wise defined function** describing the graph to the right.

Each piece should be a transformation of $y = x^2$

$$g(x) = \begin{cases} \underline{\hspace{2cm}} & \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} & \underline{\hspace{2cm}} \\ \underline{\hspace{2cm}} & \underline{\hspace{2cm}} \end{cases}$$



2). The most recent time Mr. D needed a piecewise defined function in his real life.....

I gave a computer science quiz. After looking over the problems and assigning points, it turned out there were 113 points total. I thought about where I thought the grade cut-offs should be and decided on the following mapping. [this is all true (except the last year part it may be more than one year ago now)]

In other words: 113 points should map to a 100%. 83 points should map to a 90%. 98 points (which is halfway between 83 and 113) should map to 95%.

Raw point totals	Percentages
83 – 113	90 – 100
70 – 83	80 – 90
50 – 70	70 – 80
0 – 50	0 – 70

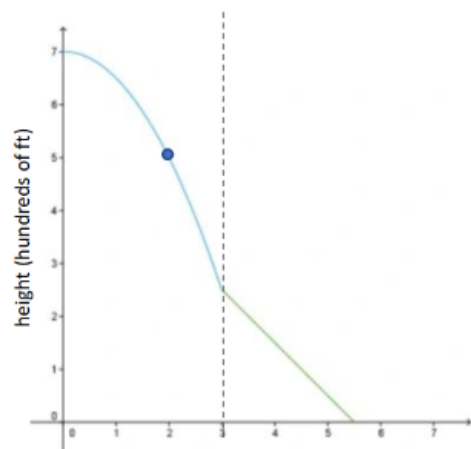
I need a function that can assign a percentage for any number of points, though. Eg. What percent should 92 points be?

Define $f(x)$ piece-wise in a way that makes sense and sketch its graph.

(btw: I actually did this myself, and had my computer use the equation to change all the raw scores into percents automatically for all ~120ish students).

3). The function $f(x)$ gives the **height** of a skydiver as a function of **time**. What if I wanted a function $g(x)$ which tells me the **time** the skydiver was at a certain **height**? Write the piecewise defined function for $g(x)$.

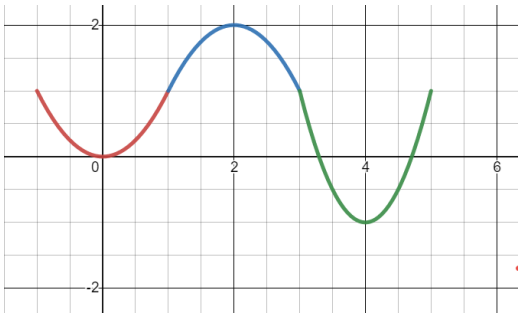
$$f(x) = \begin{cases} -\frac{1}{2}x^2 + 7, & 0 \leq x < 3 \\ -(x - 3) + 2.5, & 3 \leq x \leq 5.5 \end{cases}$$



Piecewise Defined Functions – Teaching Set 2 – Solutions

1.

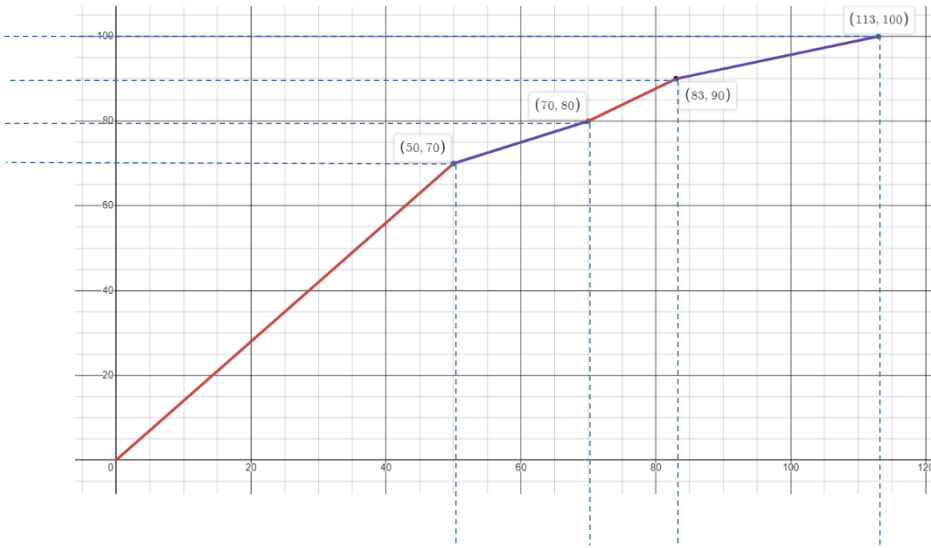
$$f(x) = \begin{cases} x^2 & -1 \leq x \leq 1 \\ -\frac{1}{2}(x-2)^2 + 2 & 1 < x \leq 3 \\ (x-4)^2 - 1 & 3 < x \leq 5 \end{cases}$$



2.

$$f(x) = \begin{cases} \frac{7}{5}x & 0 \leq x < 50 \\ \frac{1}{2}(x-50) + 70 & 50 \leq x < 70 \\ \frac{10}{13}(x-70) + 80 & 70 \leq x < 83 \\ \frac{1}{3}(x-83) + 90 & 83 \leq x \leq 113 \end{cases}$$

Raw point totals	Percentages
83 – 113	90 – 100
70 – 83	80 – 90
50 – 70	70 – 80
0 – 50	0 – 70



3.